

Part 3 – Equivalence Class Partitioning

Task 3.1 - ECP for appt_type Parameter

Class	Representative	Expected
Valid: new_patient	new_patient	Accepted
Valid: follow_up	follow_up	Accepted
Valid: urgent_care	urgent_care	Accepted
Valid: procedure	procedure	Accepted
Valid: telehealth	telehealth	Accepted
Invalid: blank	""	ValidationError
Invalid: unknown	wellness	ValidationError
Invalid: wrong case	FOLLOW_UP	ValidationError
Invalid: non-string	None	ValidationError

Task 3.2 - ECP for search_patients() Query

Class	Representative	Expected
Valid full/partial name	Ana or Ana Gomez	Result list with patient
Valid ID search	patient_id with field=id	Result list with patient
Valid no match	Nobody	Empty list
Invalid empty	""	Empty list
Invalid non-string	None or 123	Empty list
Suspicious sanitized	Ana';--	Sanitized, safe result or empty list
Invalid field	field=email	Empty list

Task 3.3 - Weak vs. Strong Analysis

For `schedule_appointment()` with 5 valid `appt_type` classes and 4 valid `duration_minutes` classes:

Weak Normal ECP needs $\max(5,4) = 5$ tests. Strong Normal ECP needs $5 \times 4 = 20$ tests.

Recommendation :

We can use Weak Normal ECP to reduce redundancy, then add a few focused strong combinations only for high-risk interactions such as `procedure` with 60 minutes or `urgent_care` with 15 minutes.